

Passivity-Based Control of Nonlinear Systems: A Tutorial

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Abstract

In this paper we survey some recent results on stabilization of nonlinear systems using a passivity approach. In the first part of the paper we treat general systems and develop a unified framework for passivity-based nonlinear control design. In the second part we center our attention on systems described by Euler-Lagrange equations, with particular emphasis on mechanical systems, power converters and AC motors.

Note This is an abridged version of the full paper which is available upon request to the authors.

1 Introduction

It is well-known that passivity properties play a vital role in designing asymptotically stabilizing controllers for nonlinear systems. At a theoretic level, a fairly complete theory has been set up for general nonlinear feedback systems – see, for instance, [12, 53, 19]. Several testing tools including nonlinear versions of the Kalman-Yacubovitch-Popov (KYP) lemma [18] are available to tell when a finite-dimensional nonlinear dynamical system is passive.

A recent breakthrough in the field of nonlinear systems is that an answer has been given to the long-standing open but fundamental question (posed by Willems [53]) of when a finite-dimensional nonlinear dynamical system can be made passive via state feedback (see [10]). More importantly, together with geometric nonlinear control theory, the results of this kind have been used systematically in [10] to unify a number of global stabilization results for interconnected nonlinear systems in the past literature. It should be mentioned that the passivity idea for stabilization was independently used for adaptive systems

in [37] and in [33] for cascaded nonlinear systems.

The purpose of this paper is to survey recent developments on passivity-based control of finite-dimensional nonlinear dynamical systems. In the first part of the paper we treat general systems and develop a unified framework for passivity-based nonlinear control design. Exploiting the particular inherent structure of physical systems, we can reasonably expect to design a stabilizing controller with better performance. In the second part, we turn our attention to the practically important class of nonlinear systems described by Euler-Lagrange (EL) equations.

2 State-space representations

We consider in this section nonlinear control systems with outputs :

$$\dot{x} = f(x) + G(x)u \quad (2.1)$$

$$y = h(x) \quad (2.2)$$

where $x \in \mathcal{R}^n$, $u, y \in \mathcal{R}^m$, $f : \mathcal{R}^n \rightarrow \mathcal{R}^n$, $G : \mathcal{R}^n \rightarrow \mathcal{R}^{n \times m}$ and $h : \mathcal{R}^n \rightarrow \mathcal{R}^m$. Assume that these functions are locally Lipschitz with $f(0) = 0$ and $h(0) = 0$.

We present here several recent results about feedback equivalence of a nonlinear system to a passive system via static state feedback or adaptive control law.¹

2.1 Feedback passive systems

Definition 1 A system (2.1)-(2.2) is said to be feedback (strictly) C^r -passive if there exists a feedback

¹We assume the reader is familiar with the basic definitions of passivity, the ubiquitous nonlinear KYP lemma [18], and the input-output and internal stability properties of EL systems as presented in, e.g. [34].

law

$$u = \alpha(x) + \beta(x)v \quad (2.3)$$

such that the system (2.1)-(2.2)-(2.3) with new input v is (strictly) C^r -passive.

Definition 2 A system $\dot{x} = f(x, \theta) + G(x, \theta)u$ with unknown parameters θ is said to be adaptively feedback passive (AFP) with a quintuple $(V, \vartheta, \tau, h, \eta)$, if there exist a positive semidefinite function V , functions ϑ and τ , all of class C^∞ , and a C^0 function h such that the resulting system with $\hat{\theta} = \tau(x, \hat{\theta}) + \bar{\tau}$ and $u = \vartheta(x, \hat{\theta}) + v$ is passive with respect to input $(v, \bar{\tau})$ and output h and the storage function $\bar{V} = V(x, \hat{\theta}) + \frac{1}{2}(\hat{\theta} - \theta)^T \Gamma^{-1}(\hat{\theta} - \theta)$. That is,

$$\dot{\bar{V}} \leq -\eta(x, \hat{\theta}) + h^T(x, \hat{\theta}, \theta) \begin{pmatrix} v \\ \bar{\tau} \end{pmatrix} \quad (2.4)$$

where η is a nonnegative function. If η is a positive definite function w.r.t. x , then system $\dot{x} = f(x, \theta) + G(x, \theta)u$ is said to be strictly AFP.

In [10], Brynes *et al.* gave a sufficient and necessary condition for the longstanding open question: when is a finite-dimensional nonlinear system made passive via state feedback? More explicitly,

Theorem 1 [10] Consider a nonlinear system (2.1)-(2.2) having a global normal form:

$$\begin{aligned} \dot{z} &= q(z, y) \\ \dot{y} &= b(z, y) + a(z, y)u \end{aligned} \quad (2.5)$$

where $a(\cdot)$ is globally invertible. Then system (2.1)-(2.2), or (2.5) is globally feedback equivalent to a (resp. strictly) C^2 -passive system with a positive definite storage function, if and only if it is globally weakly minimum-phase (resp. globally minimum-phase).

As an application of this important result to the global stabilization of cascaded nonlinear systems, it has been demonstrated in [10] that several previous stabilization schemes via different approaches can be unified. In particular, the following general result can be established.

Theorem 2 [10] Consider a cascaded nonlinear system of the form

$$\begin{aligned} \dot{\zeta} &= f_0(\zeta) + f_1(\zeta, y)y \\ \dot{x} &= f(x) + g(x)u \\ y &= h(x) \end{aligned} \quad (2.6)$$

Suppose $\dot{\zeta} = f_0(\zeta)$ is GAS at $\zeta = 0$. It is also assumed that $\{f, g, h\}$ is zero-state detectable and C^r -passive with a positive definite and proper storage function ($r \geq 1$). Then system (2.6) is GAS by smooth state feedback.

As said above, these theorems proposed in [10] are devoted to weakly minimum-phase systems with known parameters. In the sequel, we extend part of the main results in [10] to a nonlinear system with unknown parameters described by:

$$\dot{x} = \phi_0(x) + \phi(x)\theta + (\psi_0(x) + \psi(x)\theta)u \quad (2.7)$$

First of all, we give a necessary and sufficient condition for adaptive feedback passification of a nonlinear system.

Theorem 3 A system $\dot{x} = f(x, \theta) + G(x, \theta)u$ with unknown parameters θ is strictly AFP with a proper storage function V if and only if it is adaptively quadratically stabilizable².

Next, we show that the AFP property can be propagated through any minimum-phase nonlinear system with relative degree one. More specifically, consider an interconnected nonlinear system with linearly appearing parametric uncertainty:

$$\begin{aligned} \dot{\xi} &= f_{10}(\xi) + f_1(\xi)\theta + (G_{10}(\xi) + \Delta G_1(\xi)\theta)y \\ \dot{z} &= q(\xi, z, y) \\ \dot{y} &= f_{20}(x) + f_2(x)\theta + (G_{20}(x) + \Delta G_2(x)\theta)u \end{aligned} \quad (2.9)$$

with $x = (\xi^T, z^T, y^T)^T$, $\xi \in \mathcal{R}^{n_0}$, $z \in \mathcal{R}^{n-m}$ and $y, u \in \mathcal{R}^m$. Denote $G_2(x, \theta) = G_{20}(x) + \Delta G_2(x)\theta$.

Proposition 1 [20] If the ξ -subsystem of (2.9) with y considered as input possesses AFP property with a quintuple $(V_1, \vartheta_1, \tau_1, h_1, \eta_1)$, if G_2 is globally invertible for each θ , then the interconnected system (2.9) possesses AFP property with $(V_2, \vartheta_2, \tau_2, h_2, \eta_2)$.

It is worth noting that strict passivity property can be propagated under some severe conditions. In the following we give additional conditions under which AFP with adaptive regulation can be achieved for the interconnected system (2.9).

²This terminology was borrowed from [21]. A system $\dot{x} = f(x, \theta) + G(x, \theta)u$ is adaptively quadratically stabilizable if there exist a function $V(x, \theta)$ which is positive definite and proper w.r.t. x , an adaptive law $\hat{\theta} = \tau(x, \hat{\theta})$, an adaptive controller $u = \vartheta(x, \hat{\theta})$ and an augmented Lyapunov function candidate $\bar{V} = V(x, \hat{\theta}) + \frac{1}{2}(\hat{\theta} - \theta)^T \Gamma^{-1}(\hat{\theta} - \theta)$ so that

$$\dot{\bar{V}} \leq -\eta(x, \hat{\theta}) \quad (2.8)$$

where η is positive definite w.r.t. x .

Proposition 2 [20] *Under the conditions of Proposition 1, assume that ‘ $V_1(t)$ bounded’ implies ‘ $\xi(t)$ bounded’ and $\eta_1 \equiv 0$ implies ‘ $\xi(t) \rightarrow 0$ ’. If z -subsystem in (2.9) is BIBS stable and is GAS at $\xi = 0$ when $(z, y) = (0, 0)$, then the interconnected system (2.9) has AFP property with adaptive regulation.*

An iterative application of Props. 1 and 2 gives rise to the following result via integrator backstepping.

Corollary 1 [21] *Any system in strict-feedback form*

$$\begin{aligned} \dot{x}_i &= x_{i+1} + \phi_i(x_1, \dots, x_i)\theta, \quad 1 \leq i \leq n-1 \\ \dot{x}_n &= u + \phi_n(x_1, \dots, x_n)\theta \end{aligned}$$

is globally adaptively stabilizable.

3 Euler-Lagrange Systems

In this section we show how PBC can be applied to systems described by Euler-Lagrange equations, for short EL systems. We consider systems described by the EL equations of motion

$$\Sigma_p: \frac{d}{dt} \left[\frac{\partial \mathcal{L}_p(q_p, \dot{q}_p)}{\partial \dot{q}_p} \right] - \frac{\partial \mathcal{L}_p(q_p, \dot{q}_p)}{\partial q_p} = - \frac{\partial \mathcal{F}_p(\dot{q}_p)}{\partial \dot{q}_p} + M_p u \quad (3.1)$$

where $q_p \in \mathcal{R}^{n_p}$ are the generalized coordinates, $\mathcal{L}_p = T_p(q_p, \dot{q}_p) - V_p(q_p)$ is the Lagrangian. We assume $T_p(q_p, \dot{q}_p) = \frac{1}{2} \dot{q}_p^T D_p(q_p) \dot{q}_p$, $D_p(q_p) > 0$, $V_p(q_p) + c > 0$ and that the external forces are due to dissipation (captured by the Rayleigh function $\mathcal{F}_p(\dot{q}_p)$) and the control $M_p u$, with $u \in \mathcal{R}^{n_u}$ and M_p a real matrix³. It is important to note that the system is fully characterized by the triple $\Sigma_p = \Sigma_p\{T_p, V_p, \mathcal{F}_p\}$, which we call in the sequel the *EL parameters*, and the matrix M_p . Actually, the design philosophy that we propose centers around the idea of modifying, via the control, these EL parameters.

3.1 Stabilization with Euler-Lagrange Controllers

Output feedback stabilization problem Given an EL system of the form (3.1) with generalized coordinates $q_p \in \mathcal{R}^{n_p}$, regulated and measurable outputs $q_z = M_z q_p$, $q_y = M_y q_p$, respectively, and a desired constant reference $\bar{q}_z$. Design an output feedback controller that insures *global asymptotic stability* (GAS) of an equilibrium $(q_p, \dot{q}_p) = (\bar{q}_p, 0)$ satisfying $\bar{q}_z = M_z \bar{q}_p$.

³In subsection 2.4 we will see an example where the control does not enter linearly in the dynamics and we have other external forces.

In [34] we proposed the following approach to solve the problem. First, choose a controller that preserves *passivity* of the closed-loop, in particular use an *EL controller* with generalized coordinates $q_c \in \mathcal{R}^{n_p}$ and EL parameters $\Sigma_c\{T_c(\dot{q}_c), V_c(q_c, q_y), \mathcal{F}_c(\dot{q}_c)\}$. Notice that the potential energy of the controller, $V_c(q_c, q_y)$, depends on q_y . Second, to insure the closed-loop is also an EL system, hence passive, we define the control as

$$M_u u = - \frac{\partial V_c(q_c, q_y)}{\partial q_p} \quad (3.2)$$

In this way, the EL parameters of the closed loop are obtained by simple *addition*, that is $\Sigma_{cl}(T_p + T_c, V_p + V_c, \mathcal{F}_p + \mathcal{F}_c)$. The EL parameters of the controller must be chosen to insure *energy-shaping* and *damping injection*. That is, V_c should “dominate” V_p , in the sense that $V_p + V_c$ should have a unique minimum at the desired equilibrium. Further, the controller should inject the damping required for asymptotic stability. This is easy if $\dot{q}_p$ is *measurable*, else it has to come through the dynamic extension via a suitable choice of $\mathcal{F}_c$. It is clear that a *dissipation propagation* condition, (which is closely related with a detectability assumption [11], [24]), will be needed.

Proposition 3 [34] *Assume*

A.1 (Energy shaping) $\exists V_{c2}(q_y) : \mathcal{R}^{n_y} \rightarrow \mathcal{R}_+$ such that $V_1(q_p) \triangleq V_p(q_p) + V_{c2}(q_y)$ has a unique minimum at the desired equilibrium $\bar{q}_z = M_z \bar{q}_p$, for instance,

$$\frac{\partial V_1(q_p)}{\partial q_p} \big|_{\bar{q}_p} = 0, \quad \frac{\partial^2 V_1(q_p)}{\partial q_p^2} \geq \epsilon I > 0.$$

A.2 (Realizability) (3.2) is solvable for u , where

$$V_c(q_c, q_y) \triangleq \frac{1}{2} (q_c + A_1 q_y)^T A_2 (q_c + A_1 q_y) + V_{c2}(q_y)$$

with A_1 a full rank matrix and $A_2 > 0$,

A.3 (Dissipation propagation) ($u \equiv \text{const}$ and $\dot{q}_y \equiv 0$) $\Rightarrow \lim_{t \rightarrow \infty} M_y^T \dot{q}_p = 0$,

U.t.c., the EL controller

$$\Sigma_c\{T_c(q_c, \dot{q}_c), V_c(q_c, q_y), \mathcal{F}_c(\dot{q}_c)\}$$

where $\dot{q}_c^T \frac{\partial \mathcal{F}_c(\dot{q}_c)}{\partial \dot{q}_c} \geq \alpha \|\dot{q}_c\|^2$ for some $\alpha > 0$, solves the output feedback global stabilization problem.

The corollary below provides a theoretical justification to the standard practice of replacing the velocity measurement by its approximate differentiation.

Corollary 2 *The “dirty derivative” controller*

$$M_p u = - \text{diag} \left\{ \frac{b_i p}{p + a_i} \right\} q_y - \frac{\partial V_{c2}(q_y)}{\partial q_y}$$

with $p \triangleq \frac{d}{dt}$, $a_i, b_i > 0$, solves the output feedback global stabilization problem for EL systems verifying A.3 provided $V_{c2}(q_y)$ satisfies A.1.

3.2 Saturated Inputs

The methodology described above can be easily modified to take into account *input constraints*, that is

$$|u_{p_i}| \leq u_{p_i}^{\max}, \quad i = 1, \dots, n_u \quad (3.3)$$

The key observation is that, as it follows from (3.2), the control will be bounded if the derivative of $V_c(q_c)$ is saturated outside some specified ball, so that the control signal remains bounded. For instance, instead of quadratic functions, we could use $f(x) \triangleq \ln\{\cosh(x)\}$, which is positive definite and its derivative $f'(x) = \tanh(x)$ has the desired saturation property. The resulting controllers will then automatically incorporate a suitable saturation function.

Proposition 4 [28] *Consider the EL system (3.1) with saturated inputs (3.3), measurable output q_p , and a constant desired reference value $\bar{q}_p \in \mathbb{R}^{n_p}$. Assume the system is fully actuated, that is $M_p = I$. Choose the potential energy of the EL controller as*

$$V_c = V_{c2}(q_c, q_p) - q_p^T \frac{\partial V_p}{\partial q_p}(\bar{q}_p)$$

with

$$V_{c2}(q_c, q_p) := \sum_{i=1}^n \frac{k_{2i}}{b_i} \int_0^{(q_{c_i} + b_i q_{p_i})} \text{sat}(x_i) dx_i + k_{3i} \int_0^{q_{p_i} - \bar{q}_{p_i}} \text{sat}(x_i) dx_i$$

$b_i > 0$, k_{2i} sufficiently small and $\min_i k_{3i} > k_{3i}^{\min}$, with $k_{3i}^{\min}$ some suitable constant. Then $(\bar{q}_p, q_p, \dot{q}_c, q_c) = (0, \bar{q}_p, 0, \bar{q}_c)$ is a GAS equilibrium point of the closed loop provided⁴

$$u_{p_i}^{\max} > \left| \left(\frac{\partial V_p}{\partial q_p}(\bar{q}_p) \right)_i \right| + k_{3i}, \quad i = 1, \dots, n_p$$

3.3 Power Converters

In a series of papers starting with [44] we have advocated the use of PBC for voltage regulation of dc-to-dc converters. An averaged PWM model for these devices is given by the EL system (3.1) with generalized coordinates $q_p \triangleq [q_L^T, q_C^T]^T \in \mathbb{R}^n$ the electric charges in inductances and capacitances, respectively,

⁴Notice that the gradient of the systems potential energy is evaluated here at the desired reference.

$\mathcal{T}_p(\dot{q}_p, q_p)$ the magnetic co-energy in the inductances, $\mathcal{V}_p(q_p)$ the electric field energy of the capacitances, $\mathcal{F}_p(\dot{q}_p)$ the Rayleigh dissipation co-function due to the presence of resistances, and F_p the forcing functions due to voltage sources. For example, for the boost converter an average model has the EL parameters

$$\{\mathcal{T}_p, \mathcal{V}_p, \mathcal{F}_p\} = \left\{ \frac{1}{2} L \dot{q}_L^2, \frac{1}{2C} q_C^2, \frac{1}{2} R [v \dot{q}_L - \dot{q}_C]^2 \right\}$$

where $v \in [0, 1]$ is the duty cycle, $C, L, R, E > 0$ are the capacitance, inductance, resistance and voltage source, respectively. After the change of coordinates $z_1 = \dot{q}_L$ and $z_2 = q_C/C$ one obtains

$$D_p \dot{z} - v \mathcal{J} z + \mathcal{R}_p z = F_p \quad (3.4)$$

where

$$D_p = \begin{bmatrix} L & 0 \\ 0 & C \end{bmatrix}, \quad \mathcal{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathcal{R}_p = \begin{bmatrix} 0 & 0 \\ 0 & 1/R \end{bmatrix}$$

and $F_p = [E, 0]^T$. The control problem is to design $v \in [0, 1]$ such that $\lim_{t \rightarrow \infty} z_2 = \bar{z}_2$ with internal stability, where $\bar{z}_2 > E$ is the desired constant voltage. This is a challenging theoretical problem because the system (3.4) is nonlinear, nonminimum phase (with respect to z_2), with uncertain parameters and saturated controls.

In [44] we follow the PBC approach and propose to preserve the EL structure but modify the EL parameters to achieve the control objective. To this end, we choose the closed-loop EL parameters as

$$\{\mathcal{T}_{cl}, \mathcal{V}_{cl}, \mathcal{F}_{cl}\} = \left\{ \frac{L}{2} \tilde{z}_1^2, \frac{C}{2} \tilde{z}_2^2, \frac{R}{2} (v \tilde{z}_1 - C \tilde{z}_2)^2 + R_1 \tilde{z}_1^2 \right\}$$

where $\tilde{z} = z - z_d$, with z_d a desired value for z , which is to be defined later. Notice that we have chosen the desired EL parameters equal to the open-loop EL parameters, but in terms of the errors $\tilde{z}$. Also, we have added dissipation with $R_1 > 0$.

Proposition 5 [44] *The averaged PWM model (3.4) in closed loop with the controller*

$$\begin{aligned} C \dot{z}_{2d} &= -\frac{1}{R} z_{2d} + \frac{\tilde{z}_2^2}{RE z_{2d}} [E - R_1 (z_1 - \frac{\tilde{z}_2^2}{RE})] \\ v &= \frac{1}{z_{2d}} [E - R_1 (z_1 - \frac{\tilde{z}_2^2}{RE})] \end{aligned}$$

has an equilibrium point

$$(z_1, z_2, z_{2d}) = (\frac{\tilde{z}_2^2}{RE}, \bar{z}_2, \bar{z}_2)$$

which is asymptotically stable.

Remark

The result above has been extended in several directions. We have incorporated an adaptation mechanism that estimates the unknown load resistance,

combined PBC with sliding mode control to reduce the energy consumption, and treated the case of cascaded converters. We refer the reader to [14], where we carry out a comparative experimental study, for the references of all these works.

3.4 AC Rotating Machines

For AC rotating machines the generalized coordinates are the charges at windings q_{ei} , $i = 1, \dots, n_e$, and the rotor angular position q_m . If we assume that there is no magnetic saturation, flux λ and current $\dot{q}_e$ are related by

$$\lambda = D_e(q_m)\dot{q}_e + \mu(q_m)$$

where $D_e = D_e^T > 0$ is the inductance matrix of the windings, and μ represents flux linkages due to permanent magnets. The Lagrangian is

$$\mathcal{L}_p(\dot{q}_e, \dot{q}_m, q_m) = \frac{1}{2}\dot{q}_e^T D_e \dot{q}_e + \mu^T \dot{q}_e + \frac{1}{2}D_m \dot{q}_m^2 - V_p(q_m)$$

where $V_p(q_m)$ is the contribution of the permanent magnets to the potential energy. The equations of motion are

$$\begin{aligned} D_e \ddot{q}_e + W_1(q_m)\dot{q}_m \dot{q}_e + W_2(q_m)\dot{q}_m + R_e \dot{q}_e &= M_e u \\ D_m \ddot{q}_m - \tau(\dot{q}_e, q_m) + R_m \dot{q}_m &= -\tau_L \end{aligned} \quad (3.5)$$

where

$$W_1 \triangleq \frac{\partial D_e}{\partial q_m}, \quad W_2 \triangleq \frac{\partial \mu}{\partial q_m}, \quad R_e \triangleq \text{diag}\{R_1, \dots, R_{n_e}\}$$

τ_L is a load torque, and the *generated torque* is

$$\tau = \frac{1}{2}\dot{q}_e^T W_1 \dot{q}_e + W_2 \dot{q}_e - \frac{dV_p}{dq_m}$$

For underactuated machines, $M_e^T \triangleq [\mathcal{I}_{n_s}, 0]$, $n_s < n_e$, else $M_e = \mathcal{I}_{n_e}$.

In [31] we provide a solution to the following

Output feedback torque tracking problem

Given the motor equations (3.5) with state vector $[q_e^T, \dot{q}_e^T, q_m, \dot{q}_m]^T$, inputs $u \in \mathcal{R}^{n_s}$, regulated output q_m , measurable outputs $\dot{q}_s, q_m, \dot{q}_m$, smooth (bounded) disturbance $\tau_L(t)$, desired (bounded) torque $\bar{\tau}$ (with known bounded derivative), and desired constant flux norm $\beta > 0$. Define a class of machines (in terms of their EL parameters) such that the PBC controller ensures

$$\lim_{t \rightarrow \infty} (\tau - \bar{\tau}) = 0, \quad \lim_{t \rightarrow \infty} \|\lambda_r\| = \beta$$

with all internal signals bounded.

Proposition 6 [31] *Assume the machine (3.5) satisfies:*

A.1 (Decoupling) *The $n_r \times n_r$ -dimensional (2,2) block of the matrix $W_1 = \frac{\partial D_e}{\partial q_m}$ is zero, and the rotor components of the vector μ are independent of q_m .*

A.2 (Blondel-Park transformability) *There exists a constant matrix $U \in \mathcal{R}^{n_e \times n_e}$ solution of*

$$U D_e - D_e U = W_1, \quad R_e U = U R_e, \quad U W_2 = \frac{dW_2}{dq_m}$$

Under these conditions, there exists a dynamic output feedback PBC that solves the torque tracking problem above.

Remark

In [40] PBC was used to solve the problem of global tracking of robot manipulators which are driven by AC machines.

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